António Maria Wergikosky Nunes

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Summary

António Nunes holds a bachelor's and master's degree in Aerospace Engineering from Instituto Superior Técnico (IST), University of Lisbon. He is currently pursuing a PhD in the same field of study at the Institute for Systems and Robotics, IST, where he previously served as a research fellow, under a scholarship financed by LARSys and NEURASPACE. His master's thesis received the 2025 IEEE Portugal "Outstanding MSc Thesis Award" and was distinguished by the Department of Electrical and Computer Engineering at IST, in 2025, as one of the best of its academic year. In April 2026, he was selected by MIT Portugal to attend the "Space Week at MIT" in Boston, USA. His research interests include nonlinear control, orbital mechanics, and nonlinear optimization applied to the control of autonomous spacecraft in multi-body environments.

Work Experience

Research Fellow Feb 2025 - Sep 2025

Institute for Systems and Robotics, Instituto Superior Técnico, University of Lisbon

Research on advanced orbital station-keeping solutions and optimal trajectory design for the control and navigation of autonomous spacecraft in multi-body environments.

Funding: LARSyS FCT and NEURASPACE project.

Course Monitor May 2023 - Jun 2023

Instituto Superior Técnico, University of Lisbon

Monitor in the course of flight stability for one trimester. Responsible for student evaluation.

Education

2025 - **PhD student in Aerospace Engineering** (ongoing)

Instituto Superior Técnico, University of Lisbon

Host institution: Institute for Systems and Robotics.
Funding: LARSyS FCT and NEURASPACE project.

2022 - 2024 **Master's Degree in Aerospace Engineering** (Grade: 19/20)

Instituto Superior Técnico, University of Lisbon

Major in Space with emphasis on Control and Orbital Mechanics.
Erasmus at *Politecnico di Milano, Italy* – Space Engineering course.
Thesis: Orbital Station-Keeping in the Circular and Elliptic Restricted Three-Body Problems (Grade: 20/20)

2019 - 2022 **Bachelor's Degree in Aerospace Engineering** (Grade: 18/20)

Instituto Superior Técnico, University of Lisbon

Awards & Distinctions

Outstanding MSc Thesis Award (2025)

IEEE – Portugal Section

Outstanding Master Thesis Award (2025)

Department of Electric and Computer Engineering, Instituto Superior Técnico, University of Lisbon

Diploma of Academic Excellence (2022, 2024)
Instituto Superior Técnico, University of Lisbon

Diploma of Academic Merit (2020, 2021, 2023)
Instituto Superior Técnico, University of Lisbon

Projects

Levenberg-Marquardt algorithm for the determination of spacecraft trajectories in the Earth-Moon system Link to Demo
See [4]

A nonlinear optimization approach to the transition of spacecraft trajectories from the Earth-Moon Circular Restricted Three-Body Problem into an Earth-Moon-Sun high-fidelity ephemeris model.

Nonlinear control with saturation for low-thrust station-keeping in the Earth-Moon system Link to Demo
See [3]

A nonlinear control law for continuous station-keeping in the Earth-Moon system, formally incorporating an actuation saturation constraint, that yields almost global (uniform) exponential convergence guarantees under a high-fidelity model of the dynamics.

Grants

Research Fellowship Feb 2025 - Jun 2026

Funding: LARSyS FCT (DOI: 10.54499/LA/P/0083/2020, 10.54499/UIDP/50009/2020, and 10.54499/UIDB/50009/2020) and NEURASPACE Project.

Activities

Rocket Experiment Division 2020 - 2022
AeroTéc, Instituto Superior Técnico, University of Lisbon

Member of the Electronics and Recovery teams in the design of student-made suborbital rockets.

Aerospace Week 2021 - 2022
AeroTéc, Instituto Superior Técnico, University of Lisbon

Member of the Logistics and Development team of the largest Aerospace Eng. student event in Portugal.

Student Mentoring 2021 - 2022
Instituto Superior Técnico, University of Lisbon

Part of the student mentoring program, welcoming new students to the university facilities.

Languages

Portuguese (mother tongue) / **English** (C2 – proficient)

Skills

C / C++ / Python / Java / Matlab / Git / HTML / CSS / JavaScript

Publications

[4] Nunes, A., Brás, S., Batista, P., and Xavier, J. "Designing trajectories in the Earth–Moon system: A Levenberg–Marquardt approach". In: *Acta Astronautica* 249 (2026), pp. 337–357. DOI: 10.1016/j.actaastro.2026.07.020.

[3] Nunes, A., Brás, S., and Batista, P. "Nonlinear backstepping with saturation for low-thrust station-keeping of libration point orbits". In: *Acta Astronautica* 248 (2026), pp. 324–342. DOI: 10.1016/j.actaastro.2026.06.004.

[2] Nunes, A., Brás, S., and Batista, P. "A Floquet Mode LQR for Orbital Station-Keeping in Cislunar Space". In: *2026 European Control Conference (ECC)*. Presented. Reyjkjavik, Iceland, 2026, pp. tbd.

[1] Nunes, A., Batista, P., and Brás, S. "Orbital Station-Keeping in the Earth-Moon System via Nonlinear Backstepping". In: *2025 IEEE 19th International Conference on Control & Automation (ICCA)*. Tallinn, Estonia, 2025, pp. 75–80. DOI: 10.1109/icca65672.2025.11129781.